

Bootstrap 5

Beginner Friendly

Live Demos

Bootstrap for Beginners

Grid System · Rows & Columns · Text Styling · Mobile-First — all in one place, with live examples you can see right now.

CHAPTER 00

What is Bootstrap?

Think of it as a ready-made toolkit for building websites.



Simple Definition

Bootstrap is a **CSS + JS library** made by Twitter. Instead of writing all your layout and style code from scratch, you add Bootstrap's classes to your HTML and it handles the hard work for you.

You link it in your `<head>` — that's it. No installation needed:

// Put this inside `<head>`

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"/>
```



Key idea: Bootstrap gives you pre-written class names. You just add those classes to your HTML tags and the styling is applied automatically.

The 12-Column Grid System

Bootstrap invisibly divides every row into 12 equal columns. You decide how many each element takes.

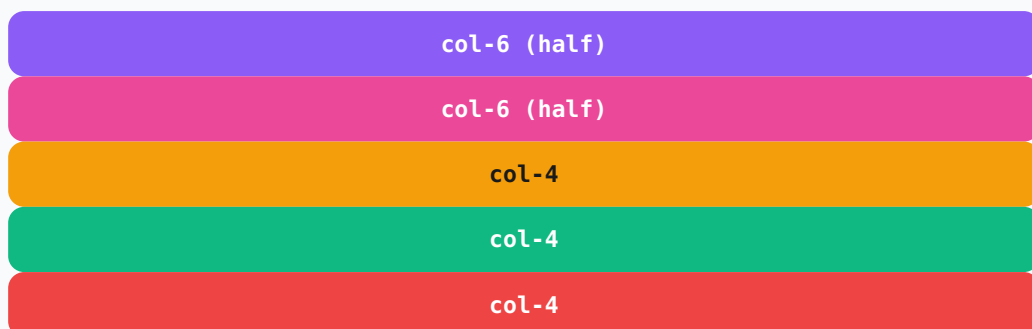
The Rule: Every row = 12 columns

You can split those 12 columns any way you like. **col-6** takes 6 of 12 (half width). **col-4** takes 4 of 12 (one-third). **col-12** takes all 12 (full width).

► LIVE DEMO – THE 12-COLUMN RULER



↑ 12 columns, each is col-1



col-8 (two-thirds)

col-4

col-12 (full width)

// The code that made the col-6 + col-6 row above

```
<!-- Every layout needs .container → .row → .col-* -->  
  
<div class="container">  
  <div class="row">  
    <div class="col-6">Left Half</div>  
    <div class="col-6">Right Half</div>  
  </div>  
</div>
```



Golden Rule: Column numbers inside one row must add up to 12 (or less). If they exceed 12, the extra column wraps to the next line.

Rows & Columns

The three-layer sandwich: container → row → col

`.container`

Centers everything and sets a max-width. Always the outermost wrapper.

`.container-fluid`

Same as container but stretches 100% of the screen. Good for hero sections.

`.row`

Creates a horizontal group of columns. Always goes inside a container.

`.col-*`

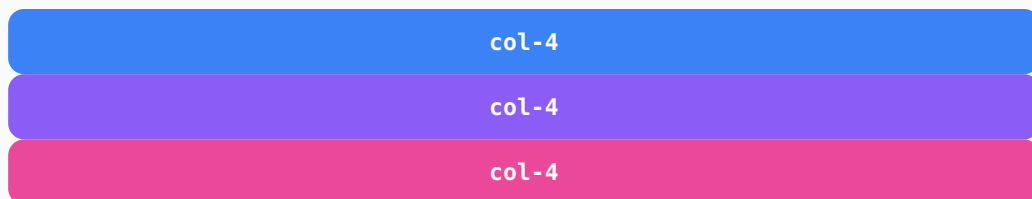
The actual column. Always goes inside a row. The number tells Bootstrap how many of 12 it takes.

Gutters (gaps between columns)

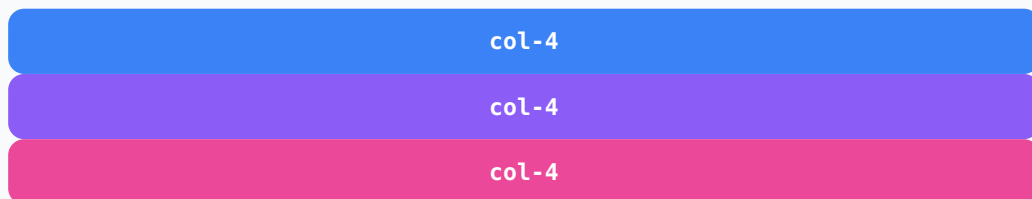
Add `g-*`, `gx-*` (horizontal), or `gy-*` (vertical) to the **.row** to add spacing between columns.

► LIVE DEMO – G-0 VS G-3 VS G-5

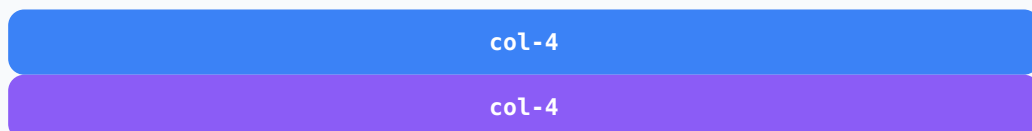
g-0 (no gap):



g-3 (medium gap):



g-5 (large gap):



col-4

// Gutter code

```
<div class="row g-3"> <!-- g-3 adds gap on all sides -->
  <div class="col-4">...</div>
  <div class="col-4">...</div>
  <div class="col-4">...</div>
</div>
```

Vertical & Horizontal Alignment

► LIVE DEMO – ALIGN-ITEMS & JUSTIFY-CONTENT

center

center

center

↑ align-items-center — columns centred vertically inside the row

col-3

col-3

↑ justify-content-center — two col-3s centred horizontally

```
<div class="row align-items-center"> <!-- vertical center -->
<div class="row justify-content-center"> <!-- horizontal center -->
<div class="row justify-content-between"> <!-- spread apart -->
<div class="row align-items-end"> <!-- stick to bottom -->
```

Breakpoints

The secret behind responsive design — Bootstrap watches the screen width and applies different column sizes.



What is a breakpoint?

A breakpoint is a screen-width threshold. When the browser crosses that width, Bootstrap can switch your layout. You control this by adding a size prefix to your col class: `col-sm-6`, `col-md-4`, etc.

Prefix	Screen Width	Device	Class Example
xs (none)	< 576px	Small phone	<code>col-12</code>
sm	≥ 576px	Large phone	<code>col-sm-6</code>
md	≥ 768px	Tablet	<code>col-md-4</code>
lg	≥ 992px	Laptop	<code>col-lg-3</code>
xl	≥ 1200px	Desktop	<code>col-xl-2</code>



You can stack multiple breakpoints on one div! Example: `col-12 col-sm-6 col-lg-4` means: full width on mobile, half on tablet, one-third on desktop.

► LIVE DEMO — RESIZE YOUR BROWSER WINDOW TO SEE COLUMNS CHANGE!

<code>col-12 col-sm-6 col-lg-4</code>
<code>col-12 col-sm-6 col-lg-4</code>
<code>col-12 col-sm-6 col-lg-4</code>
<code>col-12 col-sm-6 col-lg-4</code>
<code>col-12 col-sm-6 col-lg-4</code>
<code>col-12 col-sm-6 col-lg-4</code>



Mobile: 1 column



sm: 2 columns



lg: 3 columns

<!-- Each card:

- col-12 → full width on tiny phones*
- col-sm-6 → 2 per row on phones $\geq 576\text{px}$*
- col-lg-4 → 3 per row on laptops $\geq 992\text{px}$ -->*

`<div class="row g-2">`

`<div class="col-12 col-sm-6 col-lg-4">Card 1</div>`

`<div class="col-12 col-sm-6 col-lg-4">Card 2</div>`

`<div class="col-12 col-sm-6 col-lg-4">Card 3</div>`

`</div>`

Advanced Grid Techniques

Offsets, nesting, and auto columns — once you know these, you can build any layout.

① Offset — push a column to the right

`offset-md-2` means "skip 2 columns before starting". Useful for centering or indenting.

► LIVE DEMO — OFFSETS

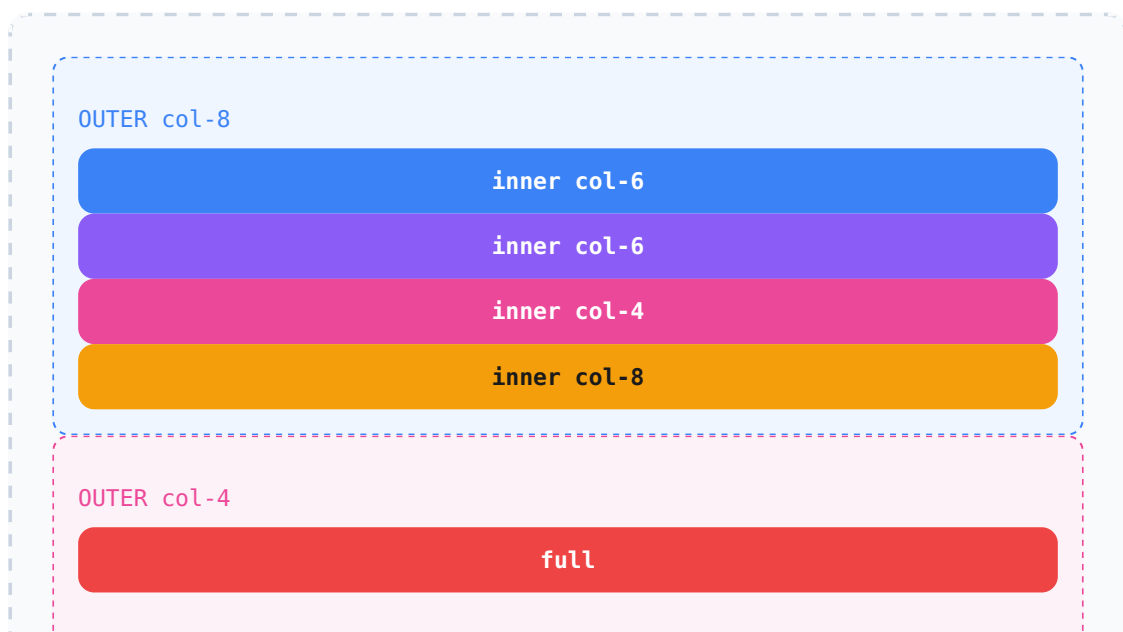


```
<div class="col-6 offset-3">Centered block</div>
<!-- offset-3 skips 3 cols, col-6 takes 6 → 3+6+3 = 12 (centred!) -->
```

② Nested Grid — a row inside a column

You can put a `.row` inside a `.col-*`. The inner row again has 12 columns, but only within the parent column's width.

► LIVE DEMO — NESTED GRID



full

```
<div class="row">
  <div class="col-8">
    <!-- Nested row - has its OWN 12 cols -->
    <div class="row">
      <div class="col-6">inner left</div>
      <div class="col-6">inner right</div>
    </div>
  </div>
  <div class="col-4">sidebar</div>
</div>
```

③ Auto Columns — let Bootstrap decide

col (no number) means "share the available space equally". Great when you don't care about exact widths.

► LIVE DEMO — AUTO EQUAL COLUMNS



↑ 4 × col — Bootstrap divides equally (each gets 3 of 12)

```
<div class="row">
  <div class="col">A</div> <!-- each gets equal share -->
  <div class="col">B</div>
  <div class="col">C</div>
</div>
```

Text & Font Styling

Bootstrap has dozens of ready-made text utility classes — no extra CSS needed.

① Display Headings — for big hero text

display-1
display-3
display-5
display-6

```
<h1 class="display-1">Huge heading</h1>  
<h2 class="display-3">Section title</h2>
```

② Font Weight & Style

fw-bold

Bold text

fw-semibold

Semi-bold text

fw-light

Light text

fst-italic

Italic text

③ Text Colours

text-primary

Primary blue

text-success

Success green

`text-danger`

Danger red

`text-muted`

Muted grey

`text-warning`

Warning yellow

`text-dark`

Dark text

④ Text Alignment

`text-start`

Left aligned

`text-center`

Centered

`text-end`

Right aligned

`text-md-center`

Left on mobile, center on md+

⑤ Size, Lead & Special Classes

This is `.lead` — a slightly larger intro paragraph

This is `.small` — smaller supporting text

`text-uppercase`

TEXT-LOWERCASE MAKES IT LOWER

`text-truncate` cuts off long text with an ellipsis automatically

This is a `.blockquote` — great for quotes.

Someone famous, *Some Source*

```
<p class="lead">Big intro text</p>
```

```
<p class="small text-muted">Small grey caption</p>
```

```
<p class="fw-bold text-primary text-uppercase">BOLD BLUE CAPS</p>
```

```
<p class="text-center text-md-start">Responsive align</p>
```

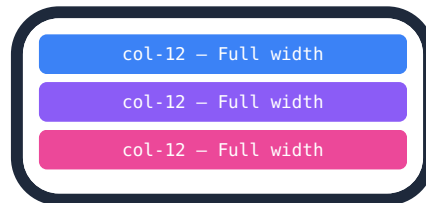
Mobile-First Approach

Design for the smallest screen first, then add styles for larger screens. Bootstrap is built this way.

What does "Mobile-First" mean?

You write the default styles for small phones first. Then you use breakpoints (sm, md, lg) to **override** those styles only when the screen gets bigger. Bootstrap's grid does this automatically when you use `col-12 col-md-6`.

 Mobile (`col-12` = stacked)



screen gets wider

 Desktop (`col-md-4` = 3 columns)



The Mobile-First Rule: Breakpoint classes apply at that width AND ABOVE. So `col-md-6` means "6-column width on *medium screens and everything wider*." On smaller screens it defaults to full width (`col-12` behaviour) unless you specify otherwise.

How to read a responsive class

```
<!-- Reading: col-12 col-sm-6 col-md-4 col-lg-3 -->
```

```
col-12    → default (phone): 1 column per row (full width)
```

```
col-sm-6  → ≥576px (phablet): 2 columns per row
```

```
col-md-4  → ≥768px (tablet): 3 columns per row
```

```
col-lg-3  → ≥992px (laptop): 4 columns per row
```

```
<div class="col-12 col-sm-6 col-md-4 col-lg-3">
  I adapt to every screen size!
</div>
```

Responsive Show/Hide Utilities

You can show or hide elements at specific breakpoints using display utilities:

```
<!-- Only visible on mobile (hidden on md and up) -->
<div class="d-block d-md-none">Mobile only</div>

<!-- Only visible on desktop (hidden below md) -->
<div class="d-none d-md-block">Desktop only</div>

<!-- Text aligns left on mobile, center on md+ -->
<p class="text-start text-md-center">...</p>
```



Always include this in your <head>! Without it, mobile browsers zoom out and Bootstrap's breakpoints won't work correctly:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
```

Quick Quiz

Test yourself — click an answer for each question.

Q1. How many columns does Bootstrap's grid system have?

6

12

16

24

Q2. What class makes text bold in Bootstrap?

text-bold

font-bold

fw-bold

bold

Q3. What does `col-12 col-md-6` mean?

Full width always

Full width on mobile, half width on tablets and above

Half width on mobile, full on desktop

Only works on tablet

Q4. Which class centres a column horizontally inside a row?

align-items-center

justify-content-center

text-center

mx-auto

Q5. What does "mobile-first" mean in Bootstrap?

You must build a mobile app first

Desktop layout is the default

Default styles target small screens; breakpoints add styles for larger screens

Bootstrap only works on mobile



Quick Cheat Sheet

Everything on one card — save or screenshot this.

GRID

```
col-1 to col-12  fixed columns
col              auto equal
col-sm/md/lg-*  responsive
offset-md-2      skip columns
g-1 to g-5       gutters
row inside col   nested grid
```

TEXT

```
display-1 to display-6  big headings
lead                    intro paragraph
small                   smaller text
fw-bold/semibold/light  weight
text-primary/muted/...  colour
text-center/start/end   align
text-uppercase           caps
fst-italic                italic
```

BREAKPOINTS

```
xs  (none) <576px  – default (mobile)
sm  ≥576px         – large phones
md  ≥768px         – tablets
lg  ≥992px         – laptops
xl  ≥1200px        – desktops
```

```
Pattern: col-12 col-sm-6 col-md-4 col-lg-3
         ↑mobile ↑phablet ↑tablet ↑laptop
```